

Business Problem Document

Retail Customer Shopping Behavior Analysis

1. Business Background

The company operates as a multichannel retailer, selling products through both online platforms and physical store locations. Recent shifts in customer purchasing behavior have been observed across different age groups, product categories, and sales channels. To remain competitive in a rapidly evolving retail environment, the company aims to better understand these behavioral changes and leverage data-driven insights to improve sales performance, enhance customer satisfaction, and increase repeat purchases.

2. Business Problem Statement

Customer purchasing decisions may be influenced by several factors, including discounts, customer reviews, seasonal trends, and payment methods. However, the company currently lacks clarity on which of these factors have the greatest impact on customer behavior. Without this understanding, marketing strategies, inventory planning, and product offerings may not effectively align with customer preferences, potentially resulting in missed revenue opportunities and reduced customer engagement.

3. Business Objective

The objective of this project is to analyze customer purchasing data to identify key trends and factors that influence buying decisions. The insights generated from this analysis will support data-driven recommendations aimed at improving marketing strategies, optimizing product offerings, and strengthening customer engagement and retention.

4. Key Business Question

How can customer purchasing data be used to identify behavioral trends, increase customer engagement, and improve overall sales strategies across both online and in-store channels?

5. Project Deliverables

5.1 Data Preparation (Python)

- Clean, transform, and organize customer purchasing data for analysis
- Address missing values, outliers, and data inconsistencies

5.2 Data Analysis (SQL)

- Analyze customer behavior, purchasing patterns, and loyalty trends
- Segment customers by demographics, sales channel, and purchase behavior

5.3 Data Visualization (Power BI)

- Develop an interactive dashboard highlighting key trends and insights
- Visualize metrics such as sales performance, customer segments, and product trends

5.4 Report & Presentation

- Summarize analytical findings in a clear and concise report
- Provide actionable recommendations to support marketing, sales, and customer engagement strategies